

ClientHello arrives
6 fragments; the sender's address proves nothing

Reassembly
128 partial messages; a full table drops its oldest

The same hello within 10 s?
yes: the stored response is sent again

CPU budget left?
a quarter of one core for failed checks; empty: dropped

UAV known, nonce new, time within 120 s?
no: refused, at no cost

ML-DSA-87 verify (pinned key): 0.08 ms, 0.45 ms on a Pi
fails: refused, and one unit of the budget is used

ML-KEM-1024 **X25519** **ML-DSA-87** **ServerHello**
encapsulate, derive, sign; 64 half-open sessions at most

a genuine hello never uses the budget: only checks that fail do

without public-key work

public-key work